MTH 451/551
Lecture 1

Prof. W. Pazner

0 What is numerical analysis?

This is a course on **numerical analysis** (specifically **numerical calculus**, which focuses specifically on calculus topics such as integration and differentiation). **Numerical analysis** is concerned with finding (or approximating) solutions to problems **numerically**, rather than **symbolically**. To give a simple example that illustrates this distinction, we could ask:

What are the roots of this polynomial?

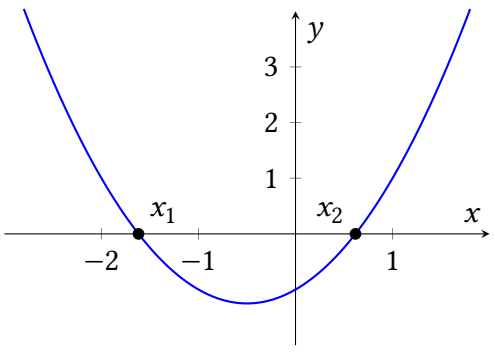
$$x^2 + x - 1 = 0$$

We could solve this problem "symbolically" or "analytically" using standard techniques such as the quadratic formula or completing the square. We would find that the solution is

$$x = \frac{-1 \pm \sqrt{5}}{2}. \tag{1}$$

We could then use, for example, a calculator to see that the roots are approximately

$$x_1 \approx -1.61803\ldots, \qquad x_2 \approx 0.618034\ldots. \tag{2}$$



Numerical analysis takes a different approach to questions like these. There are many questions in mathematics, physics, and engineering that arise quite naturally, and that **do not admit a closed-form solution**. This means that we could not write a formula for the exact roots as in (1). In fact, it is a famous theorem that a general formula (in terms of basic arithmetic operations and roots) for the roots of polynomials of degree five or higher **does not exist**. So, instead of searching for the exact solution, we try to find the approximate solution (2) directly. It turns out that there are some very good **computer algorithms** for finding these approximate solutions for very many mathematics problems, and that these algorithms give us solutions to many problems for which finding a symbolic solution would be impossible. **Numerical analysis** transforms large classes of problems from "come up with a clever substitution or apply complicated transformations" to "run a deterministic computer algorithm". The utility of such methods is enormous in many fields of science and engineering.

Numerical analysis is about **developing** and **understanding** computer algorithms to **numerically solve** mathematics problems of relevance to science and engineering.

1 Polynomial Approximation and Taylor's Theorem

[Textbook reference: §1.1]

As described above, the goal of numerical calculus is to **numerically approximate** the solution to calculus problems, such as computing derivatives or integrals of given functions. One of the most useful tools to this end will be **polynomial approximation**. We know how to differentiate and integrate polynomials (just apply the power rule). So, if, given a function $f(x)$, we can find a polynomial $p(x)$ that is *very close* to $f(x)$ (in some sense of "*close*"), then differentiating or integrating $p(x)$ could give us a good approximation of the derivatives or integrals of $f(x)$. Here, differentiation and integration are essentially stand-ins for many other operations that we might want to perform on $f(x)$, and which we know how to do for polynomials like $p(x)$. Recall that a polynomial is a function $p : \mathbb{R} \to \mathbb{R}$ given by

$$p(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n = \sum_{k=0}^{n} a_k x^k.$$

The largest number n such that $a_n \neq 0$ is called the **degree** of the polynomial.

It turns out that (under some conditions), functions (and their derivatives and integrals) can be approximated *very well* by polynomials. The key result is called **Taylor's Theorem**, which we will state shortly. We consider a real-valued function $f : [a, b] \to \mathbb{R}$. Here, $[a, b]$ is the *domain* of f, which just means that it's the interval on which the function is defined. We allow $a = -\infty$ and $b = \infty$, so depending on the function, f may be defined on all of $\mathbb{R}$. We are interested in **continuous functions**. A function is continuous at x_0 if

$$\lim_{x \to x_0} f(x) = f(x_0).$$

A function is continuous on $[a, b]$ if it is continuous at x_0 for all $x_0 \in [a, b]$. This means that $f(x)$ doesn't have any "jumps" or "gaps". Its graph can be drawn without lifting the pencil off of the page. We also assume that f is **differentiable**, meaning that its derivative exists. We use the notation

$$f'(x) = \frac{df}{dx}(x).$$

If $f'(x)$ is also differentiable, then we say f is twice differentiable, and write

$$f''(x) = f^{(2)}(x) = \frac{d^2 f}{dx^2}(x).$$

Generalizing, f may be n times differentiable, with nth derivative $f^{(n)}(x) = \frac{d^n f}{dx^n}(x)$. We usually want f to be n times **continuously differentiable**, meaning that, not only does $f^{(n)}$ exist, but that $f^{(n)}$ is also continuous. For example, we would say that f is continuously differentiable if $f'(x)$ exists and is continuous. To express that f is n times continuously differentiable, we write

$$f \in C^n[a, b].$$

We are now ready to state Taylor's Theorem.

Theorem 1 (Taylor's Theorem)**.** *Suppose that f is n times differentiable at c. There is a unique polynomial p of degree $\leq n$ for which $p^{(k)}(c) = f^{(k)}(c)$ for $0 \leq k \leq n$, and it can be expressed as*

$$p(x) = \sum_{k=0}^{n} \frac{f^{(k)}(c)}{k!}(x - c)^k. \tag{3}$$

Now suppose that $f \in C^{n+1}[a, b]$ and $c \in [a, b]$. Then for each $x \in [a, b]$ there is a $\xi = \xi(x)$ between x and c such that

$$f(x) - p(x) = \frac{f^{(n+1)}(\xi(x))}{(n+1)!}(x - c)^{n+1} \quad \text{for all } x \in [a, b]. \tag{4}$$

Furthermore, $\xi(x)$ may be chosen so that $f^{(n+1)}(\xi(x))$ is continuous on $[a, b]$.

The polynomial p is called the **Taylor polynomial of order n for f at c.** The degree of p is at most n (but it may sometimes be less than n, so "degree" and "order" may be different).

Example 1 (Approximation by Taylor polynomials)**.** Let $f(x) = \ln(1 + x)$, which is infinitely differentiable on $(-1, \infty)$. The derivatives of f can be computed:

$$f^{(n)}(x) = \frac{(-1)^{n-1}(n-1)!}{(1+x)^n} \quad \text{for } n \geq 1.$$

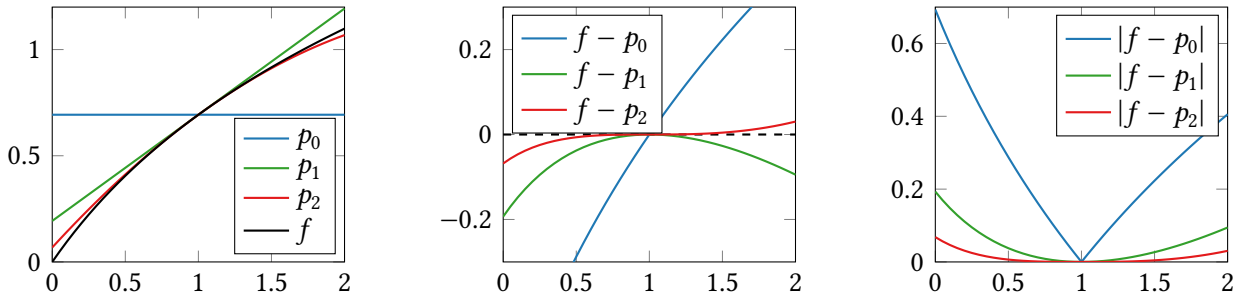
The Taylor polynomials of order n for f at $c = 1$, for $n = 0$, $n = 1$, and $n = 2$, are

$$p_0(x) = \ln(2), \qquad p_1(x) = \ln(2) + \frac{x-1}{2}, \qquad p_2(x) = \ln(2) + \frac{x-1}{2} - \frac{(x-1)^2}{8}.$$

Using (4) with $n = 2$, we see that, for example,

$$f(x) - p_2(x) = \frac{2/(1+\xi)^3}{6}(x-1)^3$$

for some $\xi = \xi(x)$ between x and $c = 1$. This gives us an expression for the **error**. It tells us how far the polynomial approximation $p_2(x)$ can be from $f(x)$. Graphs of these polynomials over $[0, 2]$ are pictured below, together with f, as well as the errors $f - p_n$ and the absolute errors $|f - p_n|$.



The plots above show Taylor polynomials of order 0, 1, and 2, centered at $c = 1$, for the function $f(x) = \ln(1 + x)$ (left), as well as errors (center) and absolute errors (right). ♦

In the next example, we will use the following theorem.

Theorem 2 (Integral Mean Value Theorem)**.** *Let $f, g \in C(a, b)$, with f bounded on (a, b) and $\int_a^b |g(x)|\,dx < \infty$. If g does not change sign on (a, b), then there is an $\eta \in (a, b)$ such that*

$$\int_a^b f(x)g(x)\,dx = f(\eta)\int_a^b g(x)\,dx.$$

Example 2 (The midpoint rule for approximating $\int_a^b f(x)\,dx$)**.** Suppose that $f \in C^2[a, b]$ and $c = (a + b)/2$. Let $p(x) = f(c) + f'(c)(x - c)$. This is the first-order Taylor polynomial of f at the midpoint of $[a, b]$. We call this function the *linearization of f at c*, since $p(x)$ is a linear approximation to $f(x)$. Using (4), we obtain

$$\int_a^b f(x)\,dx = \int_a^b p(x)\,dx + \frac{1}{2}\int_a^b f''(\xi(x))(x-c)^2\,dx.$$

We can compute

$$\int_a^b p(x)\,dx = \int_a^b f(c)\,dx + \int_a^b f'(c)(x-c)\,dx.$$

The first term is given by

$$\int_a^b f(c)\,dx = f(c)(b-a).$$

The second term is

$$\begin{aligned}\int_a^b f'(c)(x-c)\,dx &= f'(c)\,\frac{x^2}{2}\bigg|_a^b - f'(c)c(b-a)\\ &= f'(c)\frac{b^2-a^2}{2} - f'(c)\frac{b^2-a^2}{2} = 0.\end{aligned}$$

We could also see that this integral is zero since the function $x-c$ is antisymmetric (odd) about the midpoint of the interval $[a,b]$.

Using the Integral Mean Value Theorem (Theorem 2), we can compute

$$\begin{aligned}\frac{1}{2}\int_a^b f''(\xi(x))(x-c)^2\,dx &= \frac{1}{2}f''(z)\int_a^b (x-c)^2\,dx\\ &= \frac{1}{2}f''(z)\,\frac{1}{3}(x-c)^3\bigg|_a^b\\ &= \frac{1}{24}f''(z)(b-a)^3,\end{aligned}$$

for some $z = \xi(\eta) \in (a,b)$. This gives the approximation

$$\int_a^b f(x)\,dx = f(c)(b-a) + \frac{1}{24}f''(z)(b-a)^3.$$

The quantity $f(c)(b-a)$ is the basic *midpoint rule* approximation of $\int_a^b f(x)\,dx$. If the interval we're integrating over is very small, then the error will also be small (as long as the second derivative of f is not too large). ♦

> The midpoint rule for approximating integrals is the first numerical analysis algorithm we have seen in this class. If we want to approximate the integral of $f(x)$ over the interval $[a,b]$, then the midpoint rule gives us the approximation $\int_a^b f(x)\,dx \approx f(c)(b-a)$, which can be computed easily as long as we have some way to evaluate f. This is an example of **numerical quadrature**, which is the term used for methods to approximate integrals. We will discuss much more about quadrature later in this course.

In the above example, the error scaled like $(b-a)^3$. The width of the interval is $b-a$, and we considered what happens as the intervals become smaller. We often use the symbol h to denote a small parameter, such as the interval width. That makes the following form of Taylor's Theorem often convenient.

Theorem 3 (Taylor's Theorem with h)**.** *Suppose that $f \in C^{n+1}[c-\delta, c+\delta]$ for some $\delta > 0$. For all h, $|h| \leq \delta$, there is z between c and $c+h$ such that*

$$f(c+h) = \sum_{k=0}^{n} \frac{f^{(k)}(c)}{k!} h^k + \frac{f^{(n+1)}(z)}{(n+1)!} h^{n+1}.$$

Example 3 (Difference approximations of $f'(c)$)**.** Suppose that $f \in C^2[c-\delta, c+\delta]$ for some $\delta > 0$, and let $0 < h < \delta$. Applying Taylor's Theorem with h and $n = 1$, we find that there are $y_+ \in (c, c+h)$ and $y_- \in (c-h, c)$ such that

$$f(c+h) = f(c) + f'(c)h + \frac{f''(y_+)}{2} h^2,$$
$$f(c-h) = f(c) - f'(c)h + \frac{f''(y_-)}{2} h^2.$$

Rearranging the first of these and dividing by h gives the *forward difference* approximation of $f'(c)$,

$$\frac{f(c+h) - f(c)}{h} = f'(c) + \frac{f''(y_+)}{2} h.$$

Similarly, rearranging the second and dividing by h gives the *backward difference* approximation,

$$\frac{f(c) - f(c-h)}{h} = f'(c) - \frac{f''(y_-)}{2} h.$$

In both cases, the **error scales like** h. This means that halving h (the distance between the points) gives a result that is about twice as accurate (the error is roughly half).

Instead of picking one direction, we can use the symmetric *centered difference* approximation,

$$\frac{f(c+h) - f(c-h)}{2h}.$$

To study its error, we need a higher-order Taylor polynomial. Now suppose that $f \in C^3[c-\delta, c+\delta]$. Applying Taylor's Theorem with h and $n = 2$, there are $z_+ \in (c, c+h)$ and $z_- \in (c-h, c)$ such that

$$f(c+h) = f(c) + f'(c)h + \frac{f''(c)}{2} h^2 + \frac{f'''(z_+)}{6} h^3, \tag{5}$$
$$f(c-h) = f(c) - f'(c)h + \frac{f''(c)}{2} h^2 - \frac{f'''(z_-)}{6} h^3. \tag{6}$$

Subtracting (6) from (5), the $f(c)$ and $f''(c)$ terms cancel, and dividing by $2h$ gives

$$\frac{f(c+h) - f(c-h)}{2h} = f'(c) + \frac{h^2}{6} \cdot \frac{f'''(z_+) + f'''(z_-)}{2}.$$

Now the **error scales like** h^2. Halving h reduces the error by about a **factor of four**. The centered difference uses the same number of evaluations of f as the forward and backward differences, but it is much more accurate when h is small.

This is the second example of a basic numerical method that we have seen in this course. If we have a function f that we can evaluate, then we can approximate the derivative $f'(c)$ using two evaluations of f. So far, we have seen a method to approximate integrals (quadrature; specifically, the midpoint method), and a method to approximate derivatives (difference quotients, often called *finite differences*). ⧫

Finally, we introduce some notation. Suppose y is a quantity of interest that we wish to approximate by $\hat{y}$. We write $\hat{y} \approx y$ to mean that $\hat{y}$ approximates y. The **signed error** of the approximation is $y - \hat{y}$ and the **absolute error** is $|y - \hat{y}|$. The **relative error** is $|y - \hat{y}|/|y|$ (note that this is not defined if $y = 0$). The advantage of the relative error is that it is **scale invariant**, meaning that if both y and $\hat{y}$ are scaled by the same nonzero number, then the relative error does not change, as can be seen as follows. Let $Y = cy$, and $\hat{Y} = c\hat{y}$ for $c \neq 0$. Then,

$$\frac{|Y - \hat{Y}|}{|Y|} = \frac{|cy - c\hat{y}|}{|cy|} = \frac{|c||y - \hat{y}|}{|c||y|} = \frac{|y - \hat{y}|}{|y|}.$$